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REVIEW

Research Progress of High Efficiency Magnetic Refrigeration Technology and Magnetic Materials

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Abstract

As an environmentally friendly and efficient refrigeration method, magnetic refrigeration technology has always attracted the attention of researchers. In this paper, the research and development status of magnetic refrigeration technology and magnetic materials are discussed in depth, and the challenges faced by the technology are also evaluated. Firstly, the principle of magnetic refrigeration technology is introduced, and the research and development process and current development of room-temperature and low-temperature magnetic refrigeration systems are briefly described. Then, the simulation of magnetic refrigeration technology in the current era is discussed in detail, and the important role of calculation model in optimizing system design is emphasized. Secondly, this paper reviews the classification of magnetic refrigeration materials, including room-temperature magnetic refrigeration materials and low-temperature magnetic refrigeration materials, and further analyzes the current research process of magnetic refrigeration materials. Through the comparative analysis of different materials, find out the most suitable magnetocaloric materials. Finally, this paper not only summarizes the current situation of magnetic refrigeration technology, but also makes a prospect for the future development of magnetic refrigeration technology and magnetic materials. Through the in-depth elaboration and analysis of this review paper, it aims to provide theoretical and practical guidance for the further advancement of magnetic refrigeration and magnetic materials research.

Keywords Refrigeration technology · Magnetocaloric effect · Magnetic materials · Numerical simulation

1 Introduction

Magnetic refrigeration technology is an environmentally friendly, energy-saving, and efficient refrigeration technology, which utilizes the magnetic refrigeration cycle formed by the magnetocaloric effect (MCE) of magnetic materials to achieve refrigeration. Compared with traditional technologies, magnetic refrigeration does not rely on chemical refrigerants, such as chlorofluorocarbons (CFCs) and hydrofluorocarbons (HFCs), and is an important source of global greenhouse gases. In addition, the magnetic refrigeration system is more efficient because it directly uses the change of the magnetic field to cause the temperature change of the material, rather than the compression-expansion cycle commonly used in traditional refrigeration. In addition, magnetic refrigeration systems can operate over a very wide temperature range, from near absolute zero to room temperature or higher, which makes it very useful in a variety of applications, including cryogenic scientific research and ordinary commercial or residential refrigeration. Although magnetic refrigeration technology has so many advantages, its research still faces challenges, such as selecting suitable magnetic materials, optimizing system structure and parameters, and ensuring the influence of magnetic field stability and uniformity on system performance. In 1881, E. Warburg discovered the ability of iron to exhibit the magnetocaloric effect. Subsequently, in 1907, research by P. Langevin revealed the phenomenon of temperature reduction during the adiabatic demagnetization of a paramagnetic substance. In 1926, Giauque and Debye predicted the potential for refrigeration utilizing the magnetocaloric effect. In 1933, MacDougal and Giauque conducted the first adiabatic demagnetization experiment using $Gd_2(SO_4)_{3.8}H_2O$ as

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the experimental material, successfully achieving ultra-low temperatures of 0.5–0.1 K. With the active research and development efforts of scientists and engineers, magnetic refrigeration technology gradually transitioned from low-temperature to high-temperature applications [1]. In 1976, NASA Lewis and G.V. Brown successfully conducted room-temperature magnetic refrigeration experiments using Gd as the magnetic refrigeration material, achieving success under a 7-T magnetic field. Then, in December 1996, engineer C. Zimm of a space company selected rare earth materials as the magnetic working substance and utilized water with added antifreeze as the heat transfer medium. By employing niobium-titanium superconductors to generate a magnetizing field, they successfully developed a prototype of room-temperature magnetic refrigeration [2].

In recent years, with the development of materials science, thermodynamics, and magnetism, further research and exploration of magnetic refrigeration technology have been conducted. Researchers are dedicated to discovering new types of magnetic materials, improving the design of magnetic refrigeration systems, and optimizing magnetic field control techniques to enhance cooling efficiency, expand the operating temperature range, and reduce energy consumption. Naoki Hirano [3] conducted experiments to demonstrate that repeated insertion and withdrawal of magnetic shields between the magnetic field generation source and the magnetocaloric material within the gap can induce magnetic field changes. Several different types of magnetic shielding materials, including Permalloy alloys and electrical steel plates, were used to measure temperature changes after inserting magnetic shields into the gap. The study concluded that magnetic shielding can control the magnetic force generated by superconducting coils, combining the magnetic field generated by the coils with magnetic refrigeration to efficiently cool high-temperature superconducting coils. Chdil, O [4], proposed an efficient optimization method for the cooling performance of multi-layer active magnetic regenerator (AMR) based on artificial intelligence. A validated numerical model was used to predict the temperature difference between the hot and cold sources of the active magnetic regenerator. By using nanofluids as heat transfer media, the maximum temperature range of the device was increased, resulting in an approximately 20% enhancement. Additionally, optimization of ten key parameters, including geometric parameters and operating conditions, significantly improved the thermodynamic performance of the active magnetic regenerator prototype by nearly 40%. Choi, J [5], introduced a novel testbed for room-temperature magnetic refrigeration systems (MR systems). The distinctive features of this testbed include magnetic components for serial and parallel active magnetic regenerator (AMR) modes, a simplified hydraulic system with a novel sequential flow controller, and a plug-in AMR with mechanically packaged magnetocaloric materials in its original design. Terada, N [6], proposed an efficient cooling technology using holmium (Ho) as a replacement for traditional high-magnetic-field (at least 5 T) cooling systems that require superconductors. With a small magnetic field change ($\Delta\mu_0 H \leq 0.4$ T), a cooling efficiency of $-\Delta SM/\Delta\mu_0 H = 32\ J{\cdot}kg^{-1}{\cdot}K^{-1}{\cdot}T^{-1}$ can be achieved, which is an order of magnitude higher than that obtained using typical magnetocaloric materials. The magnetization intensity of Ho undergoes sharp changes with temperature and magnetic field variations. This technology can be implemented using permanent magnets, making it a suitable alternative to traditional gas compression-based hydrogen liquefaction cooling systems.

After years of research and development, magnetic refrigeration technology has successfully expanded from room-temperature to low-temperature applications, achieving significant advancements. This paper provides a comprehensive review of the latest developments in magnetic refrigeration technology. By integrating simulations in the digital age, advancements in magnetic refrigeration, and progress in low-temperature magnetic refrigeration, this review establishes a theoretical foundation for the research and development of refrigeration technology and the exploration of refrigeration materials.

2 Magnetic Refrigeration Technology

The principle of magnetic refrigeration technology lies in the behavior of magnetic materials: when no external magnetic field is applied, the entropy of the magnetic material is relatively high. However, upon the application of a magnetic field, the entropy of the magnetic material decreases, releasing heat. Removing the external magnetic field causes the magnetic material to absorb heat from the surroundings. To achieve cooling, a magnetic refrigeration cycle is required (Fig. 1), consisting of the following steps: (1) Magnetization Heating: The magnetic material is initially cooled to a low-temperature in the absence of a magnetic field. Then, a magnetic field is applied, causing the magnetic material to magnetize and reducing its magnetic entropy. Heat is absorbed from the surroundings, resulting in an increase in the material's temperature. (2) Isothermal Heat Exchange: While maintaining the magnetic field constant, heat is transferred from the material to the external environment or another medium through a heat exchanger, lowering the material's temperature back to its initial level. (3) Demagnetization Cooling: The magnetic field is removed, causing the magnetic material's magnetic entropy to increase. Heat is released from the system, further lowering the material's temperature. (4) Isothermal Heat Exchange: In the absence of a magnetic field, heat is absorbed from the surroundings through a heat exchanger, restoring the material's temperature to its initial

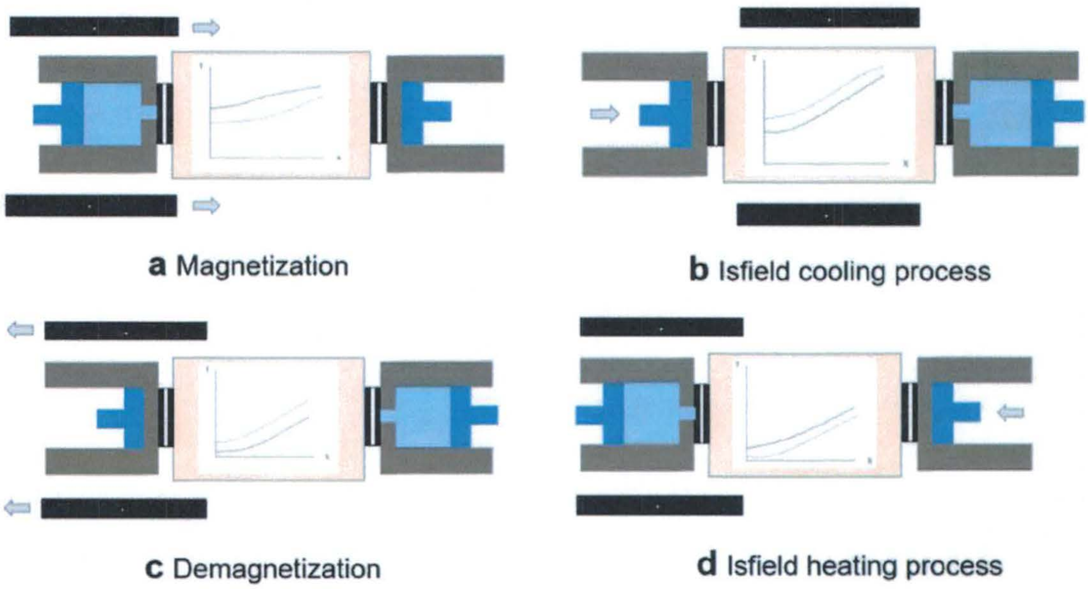

Fig. 1 Schematic diagram of refrigeration cycle

state, preparing for the next cycle [7]. A typical magnetic refrigeration system consists of a superconducting magnet as the magnetic field source, magnetic materials (such as magnetic solids or magnetic fluids), and a working substance (such as gas or liquid) [8]. Currently, magnetic refrigeration technology is mainly categorized into two types: magnetic refrigeration and low-temperature magnetic refrigeration. Room-temperature magnetic refrigeration technology is primarily focused on applications near room temperature, utilizing magnetic materials such as gadolinium (Gd) and its alloys, which typically exhibit significant magnetocaloric effects. It has been widely studied for everyday refrigeration needs, such as refrigerators and air conditioning systems. On the other hand, low-temperature magnetic refrigeration technology is applied in lower temperature ranges, such as environments below − 150 °C (approximately 123 K). This technology is mainly used for scientific research and specific industrial applications, such as cooling superconducting magnets. The materials commonly used in low-temperature magnetic refrigeration include certain rare earth elements and their compounds [9].

2.1 Cryogenic Magnetic Refrigeration Technology

The initial discovery and early experiments of the magnetocaloric effect were primarily focused on scientific research at extremely low temperatures. In 1933, American physicist William F. Giauque and his colleagues successfully conducted practical low-temperature refrigeration experiments using this effect, marking the advent of magnetic refrigeration technology. Their work concentrated on utilizing magnetic fields to alter the entropy of materials to achieve refrigeration effects in low-temperature environments [10].

Jun Shen's team [11] combined magnetic refrigeration and Giauque-MacDougall (GM) refrigeration effects by placing ErNi and TmCuAl materials in a changing magnetic field and adjusting the phase angle to optimize the integration of these two refrigeration technologies. Their experiments demonstrated that at a phase angle of 60°, the system could achieve a minimum no-load temperature of 3.5 K at a frequency of 0.5 Hz, a significant improvement over the 4.2 K of single GM refrigeration. This discovery highlights the potential of composite refrigeration technology in enhancing low-temperature refrigeration efficiency. Qiaofei Xu [12] explored the significant MCE of magnetic refrigerants across a broad temperature range, particularly in the sub-Kelvin temperature zone with a low magnetic ordering temperature (To). His research involved synthesizing and studying the performance of the new magnetic refrigerant series $Gd(OH)_{3-x}F_x$, emphasizing the adjustment of $Gd(OH)_3$'s To through the introduction of fluoride anions. Notably, when $x=2$, an excellent MCE was demonstrated within the 0.5- to 4-K temperature range, with To only at 0.5 K, marking a breakthrough in low-temperature magnetic refrigeration. Vlasov [13] proposed using paramagnetic semiconductors as active media in low-temperature adiabatic demagnetization refrigeration. The paramagnetism of impurities or structural defects can provide the entropy needed for refrigeration at low temperatures. He introduced a simple model for estimating the theoretical limits of specific entropy and cooling power achievable by demagnetization in various semiconductor systems, predicting that semiconductor paramagnetism could continue to function down to the 3-μK temperature scale. Li [14] designed an ultra-low-temperature single-stage adiabatic demagnetization refrigerator that relies on a thermal sink of about 3 K provided by a GM type refrigerator, using gadolinium gallium garnet as the magnetic refrigerant and a superconducting coil generating a maximum magnetic field of 4 T. Through the adiabatic demagnetization process, the refrigerator reached a minimum temperature of 470 mK and provided 2.7 J of cooling capacity at 1 K, with temperature fluctuations controlled within 0.5 mK. Additionally, this device provided 1.2 J of cooling capacity at 0.8 K, serving as the first stage in a three-stage adiabatic demagnetization refrigeration system aimed at reaching the 50-mK temperature zone and providing 0.7 mW of cooling power at 1 K.

These studies collectively review the latest research advancements in low-temperature magnetic refrigeration technology. Jun Shen's team demonstrated the potential of composite refrigeration technology by combining magnetic and GM refrigeration techniques and optimizing the phase angle, achieving low temperatures of 3.5 K at 0.5 Hz. Qiaofei Xu's research into chemically regulated magnetic refrigerants has enhanced their performance. Vlasov explored the potential applications of paramagnetic semiconductors in ultra-low-temperature magnetic refrigeration. Lastly, Li's development of an ultra-low-temperature single-stage adiabatic demagnetization refrigerator using superconducting coils and gadolinium gallium garnet achieved temperatures as low as 470 mK, providing an effective solution for low-temperature refrigeration. These investigations indicate that low-temperature magnetic refrigeration technology is making significant breakthroughs across multiple aspects, laying the foundation for the future development of low-temperature technologies.

2.2 Magnetic Refrigeration Technology at Room Temperature

Nebot-Andres, L. [15] found that implementing subcooling in refrigeration cycles can enhance the performance of transcritical CO_2 systems. Magnetic refrigeration devices, when used in settings with small temperature differences between the heat sink and the cooling source, can achieve higher coefficients of performance (COP) to meet the requirements for CO_2 subcooling methods. An analysis and further evaluation of subcooling methods in various environmental conditions for transcritical CO_2 cycles were conducted. Optimizations of the gas cooler pressure, subcooling degree, and operating parameters for magnetic refrigeration devices were carried out for each condition, and using existing magnetic refrigeration system prototypes for subcooling could increase the overall COP of transcritical CO_2 cycles by 9%. Ansarinasab Hojat proposed replacing traditional compression technology with AMR refrigeration technology in the precooling part of hydrogen liquefaction processes. This aims to achieve better environmental safety, reduce specific energy consumption, and enhance the isentropic efficiency of the process. Compared to traditional processes based on mixed refrigerant (MR) precooling, this could save about 5.9–15.4% of energy [16]. The team also proposed a new configuration using magnetic refrigeration precooling technology for air liquefaction cycles (Linde–Hampson, Claude, and Kapitza), which, compared to traditional air liquefaction cycles without precooling, reduced the specific energy consumption (SEC) by 11.20%, 10.96%, and 7.24% respectively, while the isentropic efficiency increased by 1.03%, 3.13%, and 2.12% respectively [17]. The Baotou Rare Earth Research Institute developed a magnetic refrigeration refrigerator using a 1.3-T NdFeB permanent magnetic field, with magnetic refrigerant being metal Gd and alloys of ϕ0.4–0.6 mm. The refrigeration space is 110 l, with a maximum refrigeration temperature difference of 22.7 ℃, reaching temperatures near 0 ℃. The temperature in the refrigerated compartment can be maintained between 5 and 10 ℃, and the external heating compensation refrigeration power is 50 W [18]. The team at the Institute of Physical and Chemical Technology of the Chinese Academy of Sciences, led by Shen Jun, researched a dual-layer rotating room-temperature magnetic refrigeration prototype. Using approximately 210 g of spherical Gd as the regenerator filler material, they achieved a refrigeration temperature span of 20.5 K, and a specific refrigeration capacity of about 47 W/kg at a 16-K temperature span [19]. Researchers at South China University of Technology, led by Wu Jianghong, utilized solid heat transfer sheets (copper sheets) to connect two sets of annular magnetic refrigerant beds positioned above and below each other. Through solid heat transfer sheets, energy transfer could be facilitated between units in the same position in the upper and lower annular beds, regardless of whether they are in the magnetized or demagnetized state. The article summarized several typical room-temperature magnetic refrigeration systems based on the development sequence of room-temperature magnetic refrigeration [20] (Table 1).

In summary, the research and application of room-temperature magnetic refrigeration technology in different fields have achieved some important results. Nebot-Andres found that the performance of transcritical CO2 can be improved by performing supercooling in the refrigeration cycle, and the magnetic refrigeration equipment can achieve high-performance coefficient when the temperature difference is small, which meets the requirements of CO2 supercooling. Ansarinasab Hojat et al. proposed the use of active magnetic regeneration refrigeration (AMR) as an alternative to

Table 1 Typical room-temperature refrigeration system

Ref	[21]	[22]	[23]	[24]	[25]	[26]	[27]	[28]	[29]
Experimental effect	The maximum temperature span is 42.28 K, and the temperature span is 18.16 K, and the cooling capacity is 51.3 W	The maximum temperature span is 25 K, and the temperature span is 18.16 K, and the cooling capacity is 44 W	The maximum refrigeration span is 16 K, and the minimum temperature is 285 K	The maximum refrigeration temperature span is 29 K, and the refrigeration capacity is 50 W when the temperature span is 10 K,COP;1.6	Maximum temperature span 22 K,cooling power between 80 and 100 W with a temperature span 22 K	Maximum cooling temperature span is 5 K	Maximum temperature span is 3.5 K, a cooling power of about 3 W under a span approaching 0 K	The maximum refrigeration temperature spans 33 K and 15 K spans 50 W	The maximum temperature span is 13.5 K,COP1.40. The maximum refrigeration temperature is 298 K
Operating frequency	0.4–1.0	0.5–4.0	~1	0–5	~0.5	< 0.25	< 0.07	0.5/0.8	0.1–1
Refrigerant interception system	Piston cylinder displacer	A pump, and valves mounted to the wheel	Pistons valve combination or a fluid displacer	A fluid displacer	The movement of the actuator and pump	A collector connected to electric valves	Two-position three-way valves	A simple piston-cylinder displacer driven	A rotary valve
Fluid form	Oscillatory flow (helium)	Compound flow (water)	Oscillatory flow (helium)	Composite flow (20% ethylene glycol solution)	Compound flow (silicone oil/water)	Composite flow (50% ethylene glycol solution)	Compound flow (water)	Composite flow (50% ethylene glycol solution)	Compound flow (water)
Magnetic working medium	1167.4 g Gd	Gd, Gd-GdEr, LaFeSiH	20.9 g Gd	110 g Gd	800 g Gd	360 g Gd	180 g Gd	420–650 g Gd	1.20 kg Gd
Source of magnetic field	Permanent magnet	Permanent magnet	Permanent magnet	Permanent magnet	High remanence permanent magnets	Permanent magnet	Permanent $Nd_2Fe_{14}B$ magnets	Permanent magnet	Permanent magnet
Magnetic field strength (T)	0–1.5	~1.5	~1.6	0–1.4	0–1.45	~1.55	0–1	0–1.54	0.01–1.25
Operation type	Reciprocating regenerator type	Rotary regenerator type	Reciprocating regenerator type	Rotary regenerator type	Reciprocating regenerator type	Reciprocating regenerator type	Reciprocating regenerator type	Rotating magnet type	Rotary regenerator type
Time	2006	2006	2010	2011	2012	2013	2013	2014	2014

Ref	[30]	[31]	[32]	[33]	[34]	[35]	[36]	[37]	[38]
Experimental effect	Maximum refrigeration temperature span of single layer Gd is 11.5 K	The temperature span of 10.2 K is 103 W, and the thermodynamic efficiency is 11.3%, a COP of 3.1	The refrigerating capacity of 15 K temperature span is 40.3 W, and that of 12-K temperature span is 56.4 W	The maximum refrigeration temperature spans 45 K and the minimum refrigeration temperature is 262 K	A cooling power of 20 W at temperature span of 12.3 K is achieved and cooling power is approximately 94 W/kg. COP is 0.3	The maximum temperature span is 1.7 K, and the maximum refrigerating capacity is 50 W	The maximum difference of SMR is 4.2 K, CMR uses Gd-Gd, the maximum temperature difference is 5.7 K, Gd-GdEr is 6.9 K	The temperature span formed was 26 K. The maximum cooling power of the magnetic refrigerator was about 100 W, and COP was 0.45	The refrigerator precooled the magnetized GGG from 4.40 to 1.94 K, and the total temperature difference was 3.85 K

Table 1 (continued)

Operating frequency	0.1–1.43	0–4	~2.5	0.1–0.4	—	—	—	0.25	0.0125
Refrigerant interception system	Mechanical exchange valves	Valve arrangements	Pistons	Mechanical exchange valves	A hydraulic piston	Mechanical exchange valves	Mechanical exchange valves	Mechanical exchange valves	–
Fluid form	Oscillatory flow	Composite flow (5% ethylene glycol solution)	Oscillatory flow (helium)	Composite flow (20% ethylene glycol solution)	Compound flow (water)	Composite flow (alkaline solution)	—	Compound flow (water)	Oscillatory flow (helium)
Magnetic working medium	Gd, PrSrMnO, LaFeCoSi	1700 g Gd-GdY	468 g Gd	100 g Gd-based material	1183 g Gd	211.7 g Gd	Gd	2000 g Gd	Gadolinium gallium garnet (GGG, Gd3Ga5O12)
Source of magnetic field	Permanent magnet	Permanent magnet	Permanent magnet	Permanent magnet	A double Halbach magnet array	Permanent magnet	Permanent magnet	Permanent magnet	The high-temperature superconducting (HTS)
Magnetic field strength (T)	~0.8	~1.33	0–1.4	~1.1	0.04–1.5	1.4	1.41	0–1.4	3
Operation type	Reciprocating magnet type	Rotating magnet type	Reciprocating magnet type	Reciprocating regenerator type	Rotary regenerator	Rotating magnet	Rotary regenerator type	Rotary regenerator type	Reciprocating regenerator type
Time	2014	2015	2016	2016	2019	2020	2022	2023	2024

the pre-cooling part in the hydrogen liquefaction process to improve environmental safety, reduce energy consumption, and improve process efficiency.

Based on the above analysis of low-temperature and room-temperature magnetic refrigeration, this paper suggests that magnetic refrigeration technology can be combined with other emerging technologies, which may bring a revolutionary breakthrough in the field of refrigeration. For example, the combination of magnetic refrigeration and thermoelectric refrigeration can create a multi-stage refrigeration system to achieve a wider temperature range and higher efficiency. The application of nanotechnology is expected to develop nanostructured materials with enhanced magnetocaloric effect. In addition, the application of artificial intelligence and machine learning algorithms to the design and optimization of magnetic refrigeration systems can significantly improve system performance and energy efficiency. By combining magnetic refrigeration with renewable energy technologies (such as solar or wind energy), environmentally friendly and efficient refrigeration solutions can be developed. These cross-domain integrations can not only overcome the limitations of a single technology, but also open up new application fields and promote the development of the entire refrigeration industry in a more sustainable and efficient direction.

3 Simulation of Magnetic Refrigeration Technology

Numerical simulation of magnetic refrigeration technology is a key research method that delves into the physical processes, optimization design, and performance enhancement of magnetic refrigeration systems through mathematical models and computational methods. With the rapid advancement of computer hardware technology and the flourishing growth in the computer software sector in recent years, computational fluid dynamics (CFD) methods based on computer technology have been widely applied in various fields. By using computers to study complex flow problems, researchers can accurately simulate the behavior of flow fields. Currently, the simulation software primarily used by researchers includes COMSOL Multiphysics and ANSYS. COMSOL Multiphysics is a widely used multiphysics simulation software that supports the coupled simulation of various physical fields, such as magnetic fields, fluid dynamics, thermodynamics, and structural mechanics [39]. In the field of magnetic refrigeration, COMSOL can be utilized to simulate the magnetocaloric effect of magnetic materials as well as the distribution of thermal flow and magnetic fields throughout the refrigeration system. ANSYS, capable of conducting simulations in structures, fluids, and electromagnetic fields [40], is used in magnetic refrigeration to analyze the thermal response of magnetic materials and magnetic field effects, as well as thermal management and optimization during the refrigeration process. In the realm of magnetic refrigeration machines, numerous research teams have achieved remarkable results, successfully conducting numerical simulations to study magnetic refrigerators. These advancements not only contribute to understanding the underlying mechanisms of magnetic refrigeration but also aid in the practical enhancement of these systems' efficiency and effectiveness.

The physical models commonly used in magnetic refrigeration simulation are laminar flow of fluid and heat transfer between fluid and solid parts of metal materials. Mathematical models generally follow the following governing equations [41]:

Continuity equation: $\nabla \cdot \vec{u} = 0$

Momentum equation: $\rho_f(\vec{u} \cdot \nabla)\vec{u} = \nabla p + \mu_f \nabla^2 \vec{u}$

Energy equation: $\rho_f C_{p,f}\left((\vec{u} \cdot \nabla)T_f\right) = k_f \nabla^2 T_f + h\beta\left(T_s - T_f\right)$

Energy equation for controlling heat transfer in solids: $k_s \nabla^2 T_s - h\beta\left(T_s - T_f\right) + Q_{MCE} = 0$

The source term Q_{MCE} represents the volume heat generated by magnetocaloric effect in MCM, and the expression is as follows: $Q_{MCE} = \rho_S C_{P,S}\left(\frac{\partial T_{ad}}{\partial H}\frac{dH}{dt} + \frac{\partial T_{ad}}{\partial T_S}\frac{dT_S}{dt}\right)$

In the equations mentioned above, the subscripts "*f*" and "*s*" represent the heat transfer fluid and the magnetic refrigerant, respectively. The variable "*u*" denotes the fluid velocity, "*p*" represents pressure, and ρ, c, k, T, and h denote density, dynamic viscosity, specific heat capacity at constant pressure, thermal conductivity, temperature, and convective heat transfer coefficient, respectively.

The simulation of magnetic refrigeration is a multi-dimensional and multi-scale complex process, which can be classified from multiple perspectives. The main classification dimensions include simulation scale (from micro to macro), simulation method (such as first-principles, finite element analysis), simulation object (material, magnetic field, heat transfer, etc.), time and space dimension, simulation accuracy, research purpose, and the number of physical fields considered. These classification methods reflect the complexity of magnetic refrigeration systems and the diversity of research. The selection of appropriate simulation methods and classification criteria depends on the specific research objectives, available resources, and required accuracy. It usually needs to be considered in multiple categories to fully understand and optimize the performance of magnetic refrigeration systems. This paper introduces several simulations from the perspective of system complexity from simple to difficult. Nielsen and colleagues employed a two-dimensional numerical model to study the performance and parameters of an active magnetic refrigerator, simulating the refrigeration process by varying the thickness of magnetic material layers, the distance between layers, operational frequency, and fluid dynamics. Through these simulations, they identified optimal utilization coefficients [42]. Nielsen et al. also conducted a comprehensive analysis of room-temperature AMR models developed before 2011 [43], thoroughly discussing how control equations, magnetocaloric effects, fluid flow, and magnetic field distribution affect the AMR cycle. Silva and others conducted a numerical analysis on a "sandwich"-style composite magnetic refrigeration structure using materials whose thermal conductivity varies with an external magnetic field as thermal control elements [44]. Their research showed that the composite structure could achieve high refrigeration power density (2.75 $W \cdot cm^{-1}$) at high operating frequencies (above 100 Hz). Lionte and team published a numerical analysis of a reciprocating active magnetic refrigerator, modeling both the regenerator's heat transfer and the flow of the heat transfer fluid. They obtained crucial parameters such as the system's COP, temperature span, and pressure drop, and proposed some optimized solutions [45]. Figure 2 shows the cooling and heating curves of the system under the conditions of an operating frequency of 0.45 Hz, a flow time fraction $F_B = 100\%$, and a utilization coefficient $U = 0.42$.

Monfared constructed single-layer and triple-layer "sandwich" models using Peltier elements as thermal switches to study the effects of thermoelectric arm thickness and current on the model's performance [46]. The research suggests that further optimization of thermoelectric cooling elements (geometric structure, current, etc.), finding efficient thermoelectric materials, and optimizing system matching (cycle, timing) could enhance the cooling performance of composite structures. These findings indicate that all-solid-state magnetic refrigeration systems have significant advantages and prospects in practical applications. J Wu developed a quasi-two-dimensional numerical simulation model to study optimal geometric parameters, operational characteristics, and system performance [47]. He proposed a novel solid-state MR system based on a new cascaded micro-unit regenerative (MUR) cycle, which allows for heat regeneration at the right time and in the appropriate spatial context. The system's heat regeneration involves unidirectional solid–solid heat conduction, minimizing the temperature difference and avoiding the heat transfer losses typically found in traditional active magnetic regenerators. The optimal rotational speed range that maximizes the specific cooling power (SCP) was determined under various configurations and operating conditions, achieving a maximum SCP range of 2.6–105.8$W \cdot kg^{-1}$. Huang. H M proposed a hybrid system using Peltier thermal switches in magnetic refrigeration to overcome the low cooling power density typical of conventional magnetic refrigerators [48]. By considering the thermal characteristics of the Peltier modules, a two-dimensional transient model was established. Compared to the pure parallel plate AMR mode, the hybrid system using Peltier thermal switches significantly improves the cooling power density at low frequencies (1.60–4.00 Hz). Additionally, compared to the pure Peltier mode, the hybrid system achieves the same cooling power with less energy consumption and a higher COP. Zheng developed a multistage magnetic refrigeration system suitable for the 20- to 80-K liquid hydrogen temperature range, featuring a three-stage AMR [49]. By establishing a two-dimensional transient AMR model, the system's performance was studied, revealing that the maximum cooling power of the three-stage AMR significantly exceeded that of a single-stage setup. Through optimizing the proportions and flow rates of various magnetocaloric materials, a maximum cooling power of 40.09 W and a thermodynamic efficiency of 28.14% were achieved, indicating the direction for multistage system design. C. Aprea [50] introduced a model to simulate the thermal behavior of AMR, which is used to predict the performance of AMR refrigerator systems. Different magnetic materials are considered, such as pure gadolinium, second-order phase transition rare earth alloys (SOMT) such as (Gd_xDy_{1-x} and Gd_xTb_{1-x}), and first-order phase transition alloys (FOMT) such as Gd_5 $(Si_xGe_{1-x})_4$ and $MnAs_{1-x}Sb_x$. Using this model, the cooling capacity, power consumption, and coefficient of performance of the cycle can be predicted. The simulation results clearly show that each magnetic material has the best operating conditions to maximize COP. The energy performance of FMT materials is generally better than that of SOMT materials, and Gd_5Si_2Ge has the best performance. The study also found that the use of a layered regeneration bed can effectively improve the COP of the system, especially for SOMT materials. $Gd_5(Si_xGe_{1-x})_4$ is the best magnetic material. Under the same working conditions, the COP is always greater than that of the traditional vapor compression device (from the minimum 40% to the maximum 62%).

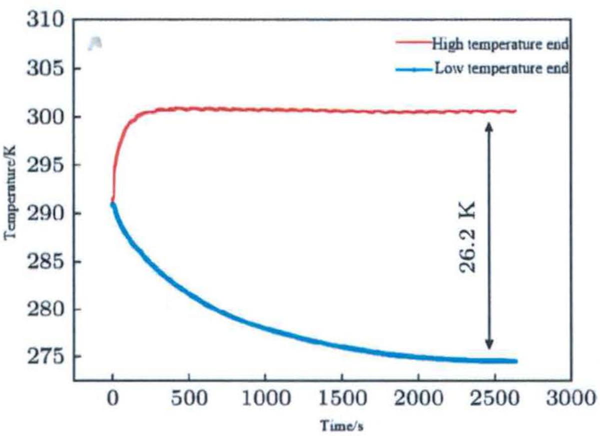

Fig. 2 Temperature curve

In conclusion, simulation plays a critical role in the research and optimization of magnetic refrigeration technology. Through simulation, researchers can thoroughly investigate and understand key factors such as thermodynamic performance, magnetic field distribution, and material characteristics of magnetic refrigeration systems. This enables the optimization of designs to enhance efficiency and reliability. Additionally, simulation can reduce experimental costs and time, accelerating the pace of technology development and deployment. It also facilitates a better understanding and prediction of the performance and effectiveness of magnetic refrigeration systems, providing guidance for system design and improvements.

4 Application of Magnetic Refrigeration Materials

After decades of research and accumulation, magnetic refrigeration technology has made some achievements in materials and systems, and found many very practical magnetic refrigeration materials, and designed a variety of high-efficiency refrigeration system structures. These research results promote the practical process of magnetic refrigeration technology. In this paper, several promising magnetocaloric materials are introduced, including Gd-based, La-Fe-Si-based, Mn-based magnetic refrigeration materials at room-temperature, multi-intermetallic compounds and rare earth metal oxides as refrigeration materials at low temperature.

4.1 Room-Temperature Magnetic Refrigeration Materials

Among many room temperature magnetic refrigeration materials, three types of materials stand out because of their unique properties and application potential: Gd metal and its alloys, La-Fe-Si compounds, and Mn-based compounds. Gd metal and its alloys are the earliest studied room temperature magnetic refrigeration materials, which are famous for their remarkable magnetocaloric effect and Curie temperature close to room temperature. La-Fe-Si compounds are favored for their giant magnetocaloric effect and adjustable operating temperature range, but they also face challenges in terms of mechanical properties and cost. Mn-based compounds, especially MnAs-based and Heusler alloys, have attracted much attention due to their excellent magnetocaloric properties and relatively low raw material costs. These three types of materials have their own characteristics, reflecting the diversity and complexity of the research on room temperature magnetic refrigeration materials. This section will explore the characteristics, advantages and disadvantages of these materials, and the latest research progress.

4.1.1 Gd Metals and Alloys

The most typical magnetic refrigeration material at room-temperature is rare earth metal Gd, and the Curie temperature $T_C = 293$ K. When the magnetic field changes 0–2 and 0–5 T, the maximum magnetic entropy changes of Gd are 5.0 and 9.7 $J\cdot kg^{-1}\cdot K^{-1}$, respectively, and the maximum adiabatic temperature changes are 7.5 K and 15 K, respectively. Gd is the first choice material for the development of room-temperature magnetic refrigeration prototype at present, and other materials are used as reference for application research.

Gadolinium (Gd) as a refrigerant requires high purity and exhibits a single phase transition temperature. To achieve a substantial MCE near room-temperature over a wide temperature range, various Gd-based rare earth alloys have been studied. However, the Curie temperatures (T_C) of most alloys drop below room-temperature without any corresponding improvement in magnetic entropy change. In 1997, Pecharsky and Gschneidner discovered the compound $Gd_5Si_2Ge_2$, which has a T_C of 274 K and exhibits a large MCE. At a magnetic field change of 0–5 T, the maximum magnetic entropy change is 18.5 $J\cdot kg^{-1}\cdot K^{-1}$, nearly twice that of Gd [51]. Rashid T. P [52] and colleagues synthesized the Gd_3Fe_4Si alloy using the arc melting method. The alloy crystallizes into hexagonal Mn_5Si_3-type Gd_5Si_3, cubic $LaNi_2$-type $GdFe_2$, and hexagonal Gd phases. The magnetization and heat capacity of Gd_3Fe_4Si alloy were measured [53]. The multiple magnetic transitions of the compound provide interesting MCE [54]. The compound's magnetization behavior at 4 K, even under a 9-T magnetic field, suggests competing ferromagnetic and antiferromagnetic interactions. The refrigerant capacity (RC) of this compound is comparable to the reference giant magnetocaloric material $Gd_5Ge_2Si_2$ and is operational at room temperature [55]. Systematic studies of Gd_3Fe_4Si alloys have revealed a new class of materials suitable for cooling applications over a wide temperature range, including room temperature [56]. Ciro Aprea [57] summarized four main solid materials for caloric refrigeration and described several main magnetic refrigeration materials for magnetic refrigeration. Gd is regarded as the reference material for magnetic refrigeration technology, showing a second-order paramagnetic to ferromagnetic transition near 294 K. $Gd_5(Si_{1-x}Ge_x)_4$ compounds, especially $Gd_5Si_2Ge_2$, show significant magnetocaloric effect near 276 K and 299 K, showing the largest adiabatic temperature change (14 K). $La(Fe_xSi_{1-x})_{13}$ alloys, such as $LaFe_{11.384}Mn_{0.356}Si_{1.26}H_{1.52}$ and $LaFe_{11.05}Co_{0.94}Si_{1.10}$, have attracted much attention due to their giant magnetocaloric effect near room temperature, and have the advantages of low cost, easy availability, and simple preparation. These comparisons provide an important reference for the material selection of magnetic refrigeration technology.

Recently, Ni-Co-Mn-In-Si-based Heusler alloys have attracted attention due to their excellent magnetocaloric properties around the second-order magnetic transition near room temperature. Venkatesan S. and colleagues investigated the effects of Gd substitution in $Ni4_5Co_5Mn_{37}In_{12}Si_1$

($Ni_{45-x}Gd_xCo_5Mn_{37}In_{12}Si_1$; $x=0$ and 5) and Gd addition ($Ni_{45}Co_5Mn_{37}In_{12}Si_1Gd_x$; $x=0$ and 5) on the thermomagnetic hysteresis behavior of the magnetization temperature and magnetic field dependency [58]. Thermomagnetic measurements indicated that Gd substitution at Ni sites significantly reduced the Curie temperature T_C from 333 to 317 K. Thermodynamic analysis of isothermal field-dependent magnetization led to a significantly larger magnetic entropy change, with $\Delta SM = -0.842$ J·kg^{-1}·K^{-1} at 305 K and an enthalpy change (ΔH) of 2 T under applied field changes. Additionally, the maximum refrigerant capacity of $Ni_2Gd_{305}Co_{40}Mn5In_5Si_{37}$ at 305 K approached 47 J·kg^{-1} [59].

Federico Scarpa explored the use of a gallium-based liquid metal—Galinstan—finding it non-corrosive to Gd at room temperature and unaffected by magnetic induction. Using this liquid metal as a heat transfer fluid in magnetic refrigeration, the observed cooling capacity increased by more than tenfold. However, due to higher friction losses, the COP was severely penalized. By adjusting the cycle time and utilization coefficient, or by changing the geometric structure using thicker sheets and pipes to achieve the same gadolinium weight, the cooling capacity was increased by 400% while maintaining the same COP [60].

In addition to the above, multi-layer magnetic materials are a promising research direction in magnetic refrigeration technology. By stacking magnetic materials with different Curie temperatures together, this material can effectively expand the operating temperature range of the refrigerator and improve the overall cooling efficiency. M. Tadout [61] investigated a 100-nm-thick Gd-Co alloy multilayer stack structure consisting of four independent Gd-Co alloy layers with different concentrations and Curie temperatures. The magnetic entropy change related to the second-order magnetic phase transition is determined by the magnetic isotherm. In addition, studies have shown that the relative cooling power (RCP) of the Gd-Co-based multilayer materials studied is higher than that of bulk Gd. In the range of 0–20kOe, the maximum magnetic entropy change (DSM) of the multi-layer alloy is 1.54 J kg^{-1} K^{-1}, which is lower than that of any single-layer Gd-Co alloy layer, but the magnetic entropy peak of the multi-layer material is wider and the RCP value is higher, reaching 200 J/kg, showing the potential of Gd-Co-based multi-layer materials in magnetic refrigeration applications. In general, although multi-layer magnetic refrigeration materials also have applications in the field of low temperature, most studies have focused on the design and optimization of multi-layer materials near room temperature due to the wider market demand and application prospects of room temperature magnetic refrigeration. However, further research is still needed to address the challenges of preparation, performance, and cost.

4.1.2 La-Fe-Si Compounds

In 1999, the Chinese Academy of Sciences discovered that compared to Gd and Tb-based compounds, La-Fe-based compounds exhibited superior magnetocaloric effects. As a result, in-depth research was conducted on these compounds [62]. The Curie temperature (T_C) of La-Fe-Si alloys can be tuned between 190 and 330 K while maintaining a significant magnetocaloric effect (ΔSM). Additionally, La is an abundant rare earth element, and Fe is highly abundant, implying low raw material costs. Therefore, La-Fe-Si-based magnetocaloric materials possess advantages such as large magnetocaloric effects, tunable Curie temperature, low cost, and non-toxic composition, making them promising candidates for commercial applications. Research results [63] indicate that the silicon (Si) content in $LaFe_{13-x}Si_x$ compounds directly influences the formation of the special crystal structure of the 1:13 phase, and appropriately increasing the silicon content can result in a significant magnetocaloric effect [64]. Researchers successfully synthesized rare earth iron-based compound $LaFe_{11.2}Co_{0.7}Si_{1.1}$ after arc melting ingots and annealing them for 30 days at 1325 K. This compound exhibited a significant magnetocaloric effect and demonstrated a strong elastic coupling phenomenon at 274 K, where the lattice underwent a significant negative expansion effect, resulting in a massive magnetocaloric effect at 274 K [65].

Since 2000, when Hu et al. discovered $La(Fe_xSi_{1-x})_{13}$ [66], researchers have extensively studied the magnetocaloric effects of this series of materials, considering it as one of the highly applicable room-temperature magnetic refrigeration materials. However, the $La(Fe_xSi_{1-x})_{13}$ materials require prolonged annealing processes to obtain higher contents of the $NaZn_{13}$ phase, although rapid solidification treatment can significantly shorten the annealing time, albeit at a high cost [67]. Recently, the team led by J. Lanzarini investigated a novel polymer extrusion molding process for two-dimensional composite structures based on $La(Fe,Si)_{13}$ alloys [68]. Through innovative techniques and meticulously designed microstructure-blade shaping tools, they successfully achieved the formation of structural blades from the powder stage to the final molding [69]. Throughout this process, the Curie temperature remained stable, with fine composites (LDPE/$La(Fe,Si)_{13}$) and structural blades exhibiting uniformity, with the highest efficiency reaching 92%. The use of atomization and powder metallurgy techniques can also shorten the heat treatment time. Performance enhancements are achieved by substituting La with R (R = Ce,Pr,Nd), Fe with Co or Mn, and by adding non-metallic elements (H,B,C) to adjust phase transition temperatures, thereby enhancing the magnetocaloric effect and reducing hysteresis losses [70].

In 2021, S.M. Wu et al. investigated the effects of $Pr_{40}Co_{60}$ on the structure, thermal conductivity, magnetism, magnetocaloric performance, and mechanical properties of sintered $LaFe_{11.6}Si_{1.4}/Pr_{40}Co_{60}$ bulk composite materials [71]. The addition of Pr-Co alloy has a significant impact on these properties due to the diffusion of Pr and Co into the $La(Fe,Si)_{13}$ phase during sintering [72]. After sintering at 1373 K for 6 h, the TC of the composite material gradually increased from 220 K to near room temperature. Under an external magnetic field change of 0–2 T, the maximum magnetocaloric entropy change $(-\Delta SM)$max ranged from 2.7 to 2.9 $J\cdot kg^{-1}\cdot K^{-1}$. The diffusion of Pr and Co atoms into the 1:13 majority phase promoted the formation of the 1:13 phase, while the Co content of the Pr-Co binder could adjust the TC of the composite material, albeit leading to a decrease in $(-\Delta SM)$max. A significant $-\Delta SM$ maximum of 2.7 $J\cdot kg^{-1}\cdot K^{-1}$ was observed under a field change of 2 T. The compressive strength of the composite material reached 459–550 MPa, with a thermal conductivity coefficient of 9.66–10.64 $W\cdot m^{-1}\cdot K^{-1}$ [73].

Eunjeong Kim [74] investigated the magnetocaloric properties of $MnxFe_{5-x}Si_3$ alloys ($x=1$, 1.1, 1.2, 1.3) for room-temperature refrigeration. By adjusting the manganese content to match the operating range and Curie temperature of magnetic refrigerants, it was found that the magnetocaloric entropy change peak and maximum adiabatic temperature change of $MnxFe_{5-x}Si_3$ decreased with increasing manganese content [75]. The study also established an internal AMR testbed to evaluate the applicability of refrigerants in AMR cycles. Using the AMR testbed with $Mn_{1.2}Fe_{3.8}Si_3$, a maximum temperature span of 4.75 K was achieved at 300 K, lower than Gd's 7.5 K. Additionally, it was found that the form of the refrigerant affected heat exchange; thus, solid magnetic refrigerants require appropriate techniques for material processing to meet the desired size and shape. Despite $MnxFe_{5-x}Si_3$ not matching Gd's magnetic refrigerant properties, it remains attractive due to material cost and processability [76].

4.1.3 Mn-Based Compounds

In 2002, Tegus et al. [77] successfully synthesized the compound $MnFe_{P0.45}As_{0.55}$, which exhibited maximum magnetocaloric entropy change of 18 $J\cdot kg^{-1}\cdot K^{-1}$ and maximum adiabatic temperature change of 9.8 K at room temperature under a magnetic field variation of 0–5 T. Subsequently, researchers extensively studied the $MnFeP_{1-x}As_x$ alloy series [78]. By adjusting the ratio of P and As, the TC of this alloy can be controlled. However, due to the toxicity of arsenic, researchers subsequently developed arsenic-free Fe_2P-type magnetic refrigeration materials, such as $Mn1.32Fe_{0.71}P_{0.5}Si_{0.56}$. The phase transition temperature of this material is approximately 270 K, exhibiting a maximum magnetocaloric entropy change of 16 $J\cdot kg^{-1}\cdot K^{-1}$ and maximum adiabatic temperature change of 2.4 K under a magnetic field variation of 2 T [79]. In 2022, Ma et al. [80] investigated the elastic thermal effect and magnetic properties of the $Ni_{50}Mn_{31.5}Ti_{18}Cu_{0.5}$ shape memory alloy. Magnetic phase transition analysis indicated that the alloy's phase transition temperature is close to room temperature. A larger unloading rate can result in smaller hysteresis and lower plastic deformation, and enhance the alloy's superplasticity. With increasing loading stress, the cooling effect of the alloy improves. When the loading stress is 600 MPa, the alloy's ΔT reaches 9 K. Moreover, the alloy exhibits good cyclic stability over 100 cycles. Therefore, the refrigeration performance of the $Ni_{50}Mn_{31.5}Ti_{18}Cu_{0.5}$ alloy can rival that of many Ni-Mn-based alloys, indicating its promising potential as an elastic refrigeration material [81].

Heusler alloys, known for their exceptional properties such as supermagnetostriction, shape memory effects, giant Hall effects, giant magnetocaloric effects, and exchange bias phenomena, have consistently been a focal point of both fundamental and applied research [82]. Miao, XJ, and colleagues selected manganese-based Heusler alloys as their test material. The samples were prepared by arc melting, followed by heat treatment, and then shaped into ribbons using melt spinning [83]. The study investigated the influence of Mn concentration on the crystal structure, magneto-thermal, and magnetocaloric properties. It was found that $Mn_{2-x}Sn_{0.5}Ga_{0.5}$ crystallizes into a hexagonal structure at room temperature. All samples underwent second-order phase transitions near room-temperature Curie points, with virtually no hysteresis. The Curie temperature and saturation magnetization of $Mn_{2-x}Sn_{0.5}Ga_{0.5}$ are highly sensitive to Mn concentration. Owing to the absence of magnetic and thermal hysteresis and a broad operating temperature range near room temperature, manganese-based Heusler alloys are seen as having significant potential in magnetic refrigeration applications [84].

Room-temperature magnetic refrigeration materials are currently a hot research topic within the field of magnetic refrigeration technology. Overall, Gd metals and alloys, La-Fe-Si compounds, and Mn-based compounds are all promising candidates for room-temperature magnetic refrigeration materials. By adjusting their composition, lattice structure, and magnetic properties, optimized magnetocaloric effects can be achieved. For Gd metals and alloys, alloying Gd with other metals can effectively enhance their magnetocaloric performance. La-Fe-Si compounds are extensively studied and applied due to their impressive refrigeration performance within a relatively low-temperature range when their composition and lattice structure are optimized. However, further research is needed to address challenges related to material stability, lifecycle, and fabrication processes to promote the widespread adoption of these materials in practical applications.

4.2 Low-Temperature Magnetic Refrigeration Materials

Low-temperature magnetic refrigeration technology shows great potential in the field of low-temperature application. Among them, rare earth element-based materials have become the focus of research on low-temperature magnetic refrigeration materials due to their unique electronic structure and strong magnetism. Among the many low-temperature magnetic refrigeration materials, the two categories of rare earth intermetallic compounds and rare earth intermetallic oxides are particularly noticeable. Rare earth intermetallic compounds, such as RAl_2, RNi_2, and RCo_2 series (R represents rare earth elements), can achieve precise temperature control in the range of 4–77 K by adjusting the type and proportion of rare earth elements. These materials usually have large magnetic entropy change and high thermal conductivity, which is beneficial to improve the refrigeration efficiency. For example, $ErAl_2$ exhibits excellent magnetocaloric properties at low temperatures, while $DyCo_2$ has received extensive attention due to its giant magnetocaloric effect.

On the other hand, rare earth metal oxides, such as R_2CuO_4 and $RMnO_3$ series, show unique advantages in extremely low-temperature (< 4 K) applications due to their unique crystal structure and magnetic phase transition characteristics. These materials usually have complex magnetic phase diagrams and rich physical properties, which provide an ideal platform for studying low-temperature magnetism and quantum phenomena. For example, Gd_2CuO_4 exhibits a significant magnetic entropy change near the liquid helium temperature, while $TbMnO_3$ is favored in low-temperature magnetic refrigeration and magnetoelectric coupling research due to its multiferroic properties. This paper mainly explores the main rare earth intermetallic compounds and rare earth intermetallic oxides, and gives suggestions.

4.2.1 Rare Earth Intermetallic Compounds

Research on low-temperature magnetic refrigeration materials primarily focuses on intermetallic compounds of rare earth metals, including binary, ternary, and quaternary compounds. Binary alloys mainly consist of RM(Ni, Ga, Pd), RM_2(Al, Co, Ni), and R_3Co compounds. Among them, RGa (R = Pr, Gd, Tb, Dy, Ho, Er, Tm) compounds possess a CrB-type crystal structure and are multiphase materials with spin reorientation (SR) and continuous magnetic transitions from ferromagnetic to paramagnetic [85]. Ternary low-temperature magnetic refrigeration materials are mainly categorized by structures such as ZrNAli-type, $MgZn_2$-type, CeFeSi-type, $ThCr_2Si_2$-type, and AlB2-type. Compounds such as R(Ni, Cu)Al, R(Ni, Cu, Pd)In, and RPbAl belong to the hexagonal ZrNiAl structure [86]. Rare earth nickel-boron alloys (RNi_2B_2C) possess a tetragonal $LuNi_2B_2C$-type structure and the RNi_2B_2C series materials exhibit coexistence of superconductivity and magnetic ordering phenomena [87]. In terms of material preparation, Yamamoto, TD, and his team investigated the crystallography, magnetism, and magnetocaloric properties of gas-atomized particles of the intermetallic compound $ErCo_2$, a giant magnetocaloric material used in low-temperature applications [88]. The experimental results indicated that atomization and heat annealing significantly altered the physical properties of $ErCo_2$. In ATO particles, the magnetic transition temperature increased from 34 to 56 K, and the phase transition shifted from first order to second order. Heat annealing shifted the transition temperature back to its original level and restored the first-order transition characteristics. The changes in magnetic properties are closely related to changes in crystallographic properties, highlighting the importance of magnetic structure coupling. The magnetic entropy change $-\Delta SM$ of the particles can be adjusted in size, shape, and peak temperature based on annealing conditions. For a magnetic field change of 0–5 T, the peak magnetic entropy change ($-\Delta SM$) varied within the range of 9–33 $J{\cdot}kg^{-1}{\cdot}K^{-1}$ [89]. This article provides some magnetic solid-state materials with excellent low-temperature magnetocaloric performance, including their magnetic transition temperatures (TM) and MCE parameters (Table 2).

4.2.2 Rare Earth Metal Oxides

In recent years, rare earth metal oxides have become a new research direction in the field of low-temperature magnetic refrigeration materials, and some materials have also been studied one after another, such as rare earth titanium oxides

Table 2 Parameters of representative rare earth intermetallic compounds

Materials	T_M (K)	$-\Delta S_M^{max}$ (J/kgK)	RC (J/kg)	Ref
PrGa	37/27	10.7	~ 140	[90]
NdGa	42/20	15.5	~ 422	[91]
ErGa	30/15	21.3	~ 512	[92]
$ErMn_2Si_2$	~ 4.5	25.2	~ 274	[93]
Er_4CoCd	12.5	20.4	392	[94]
Er_3Ni_2	17/12	19.5	~ 336	[95]
GdFeAl(0–4 T)	259	3.3	~ 261	[96]
ErFeAl	80/38	7.5	435	[97]
$GdCo_{0.5}Al_{1.5}$	70	13.2	~ 452	[98]
TbCoC	22	16.5	363	[99]
Nd_2Ni_2In	10.5	11.5	74	[100]
Dy_2Cu_2In	49.5/19.5	13.3	318	[101]
Er_4NiCd	5.9	18.3	476	[102]

Table 3 Parameters of representative rare earth metal oxides

Materials	T_M (K)	MCE ($\Delta H(T)$)	$-\Delta S_M^{max}$ (J/kgK)	Relative cooling power (RCP (J/kg))	Ref
$Yb_{0.8}Er_{0.2}MnO_3$	3.7	2	0.5	1.5	[111]
$Sm_{0.45}Pr_{0.1}Sr_{0.45}MnO_3$	132	5	7.14	258.8	[112]
$La_{0.6}Ca_{0.4}MnO_3$	260	0.5	120.2	37.4	[113]
$La_{0.7}Sr_{0.3}MnAlO_3$	367	0.01	0.162	0.388	[114]
$Pr_{0.8}Na_{0.2}K_0MnO_3$	92	5	3.2	355.6	[115]
$Zn_{0.6}Cu_{0.4}Fe_2O_4$	305	5	1.16	289	[116]
GdFeAl(0–4 T)	373	3	1.27	36.7	[117]
Er_4NiCd	664	2.5	1.39	68	[118]

[103], rare earth manganese oxides [104], and rare earth vanadium oxides [105].

Recently, advancements in the research of rare earth (RE)-based RE2TMTM′O-6 double perovskite (DP) oxides (where TM and TM′ are different three-dimensional transition metals) have been investigated by Li [106], focusing on crystal structure, magnetism, and MCE characterization. Some Gd-based DP oxides have shown promising low-temperature magnetocaloric performance. Liu synthesized polycrystalline $Eu_3B_2O_6$ using an improved solid-state reaction method, which is a novel low-temperature magnetocaloric material [107]. Precise studies on the compound's magnetocaloric effect, magnetic network properties, and crystal structure have been completed. The compound undergoes a second-order magnetic phase transition from ferromagnetic to paramagnetic state at 8 K, and the MCE was evaluated using magnetic entropy change, temperature-averaged entropy change, and $Eu_3B_2O_6$'s RC [108]. Under a field change from 0 to 5 T, the maximum values of ΔSM and TEC are 38.6 and 35.7 $J{\cdot}kg^{-1}{\cdot}K^{-1}$, respectively, with a corresponding RC reaching 435.9 $J{\cdot}kg^{-1}$. K El Maalam [109] studied the magnetocaloric properties of $(La_{0.45}Nd_{0.25})Sr_{0.3}MnO_3$ (LNSMO)-based composite materials. The structure, microstructure, magnetism, and magnetocaloric properties of LNSMO and LNSMO/5CuO samples were investigated to clarify the role of the secondary phase (CuO) in driving the magnetocaloric behavior. When the magnetic field varies from 0 to 1.5 T, the isothermal entropy change of the LNSMO/5CuO composite material is 2.55 J/kg·K, while that of the parent material LNSMO is only around 1.1 J/kg·K. This finding will provide insights and open up new avenues for enhancing the magnetocaloric effect of manganese-based materials. Pressley, LA [110], used laser diode floating zone technique to grow single crystals (diameter 4 mm, length 3 cm) of multi-component perovskite $SmCo_{5/3}Cr_{1/4}Fe_{1/4}Mn_{1/4}O_1$ to understand the interactions of "high-entropy" oxides (HEO). It deviates from the magnetic behavior of its parent $SmMO_3$ (M = Co, Cr, Fe, Mn), exhibiting two broad transitions and significant bifurcation in zero-field cooling and field cooling measurements. Measurement of magnetization intensity as a function of field displays spin canting of transition metal ions, and exchange bias appears as a function of temperature. The use of the floating zone technique for crystal growth is advantageous and provides insights for magnetic characterization (Table 3).

Rare earth intermetallic compounds and rare earth metal oxides are important materials in the field of low-temperature magnetic refrigeration. Rare earth intermetallic compounds, such as the $Gd_5(Si_xGe_{1-x})_4$ series, can control the Curie temperature of these compounds by adjusting the ratio of silicon and germanium, enabling magnetic refrigeration at different temperature ranges. On the other hand, rare earth metal oxides, such as Er_2O_3, exhibit excellent magnetic and magnetocaloric properties at low temperatures and are widely used in low-temperature magnetic refrigeration systems, especially in ultra-low-temperature ranges. However, considering aspects such as economy and production technology, achieving comprehensive applications of magnetic refrigeration materials still requires extensive research.

5 Conclusions

In this review, the development and application of magnetic refrigeration technology and the research of magnetic refrigeration materials are discussed. The following conclusions are drawn.

1. Magnetic refrigeration technology shows its potential in environmental protection, energy saving, and high efficiency refrigeration. It has the advantages of extremely low working temperature, no mechanical parts, high efficiency and energy saving, low noise and vibration, stability and accuracy, and adjustability. With the development of science and technology, we have witnessed the development of magnetic refrigeration technology from theoretical exploration, experimental verification to practical application. These make magnetic refrigeration technology have wide application prospects in various fields.

2. Simulation plays an important role in the research and development of magnetic refrigeration technology. By establishing numerical model and simulation program, researchers can accurately predict and optimize the performance and parameters of magnetic refrigeration system. Simulation technology can help optimize the structure design, working parameters, and refrigeration performance of magnetic refrigeration system, thus improving the energy efficiency and stability of the system.
3. In magnetic refrigeration technology, the application of magnetic refrigeration materials is very important to realize high-efficiency refrigeration. In terms of room-temperature refrigeration materials. By adjusting the material composition, lattice structure and magnetic properties, the researchers improved the refrigeration effect and efficiency of magnetic refrigeration materials in different temperature ranges. In the field of low-temperature magnetic refrigeration materials, rare earth intermetallic compounds, rare earth metal oxides, and other materials show excellent magnetocaloric effect and magnetic properties, and become important materials in the field of magnetic refrigeration.

This paper reasonably predicts the key points of future development: First, strengthen the exploration of new magnetic materials, especially rare earth substitution materials and multifunctional composite materials with giant magnetocaloric effect. Secondly, the system design is optimized, focusing on improving heat exchange efficiency and magnetic field utilization, while reducing system complexity and cost. Thirdly, the microscopic mechanism of magnetocaloric effect is deeply studied, and the essence of material performance optimization is revealed by advanced calculation methods and experimental techniques. Fourthly, an intelligent magnetic refrigeration system is developed, which integrates artificial intelligence and Internet of Things technology to achieve adaptive control and maximize energy efficiency. Fifth, explore the synergistic application of magnetic refrigeration and other emerging refrigeration technologies, such as thermoelectric refrigeration or adsorption refrigeration hybrid systems. Breakthroughs in these directions are expected to greatly improve the performance and economic feasibility of magnetic refrigeration and accelerate its wide application in various temperature fields.

Although magnetic refrigeration technology still faces challenges in some aspects, such as the improvement of material properties, the optimization of system design and the reduction of cost, with the continuous progress of science and technology and the continuous development of magnetic refrigeration technology, it is reasonable to believe that magnetic refrigeration will play an increasingly important role in the future and bring more innovative and sustainable refrigeration solutions to all walks of life.

Author Contribution YJ: Conceptualization and methodology (supporting), investigation, software, visualization, and writing/original draft; HP: Conceptualization and methodology, supervision, writing/review, and editing; SL: Conceptualization and methodology, supervision, writing/review, and editing.

Data Availability No datasets were generated or analysed during the current study.

Declarations

Competing Interests The authors declare no competing interests.

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